

GlucoTwin

Exploring the future of personalised glucose-risk prediction.

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PRECISION / PREVENTION / PERSONALISATION

THE CHALLENGE

Glucose risk is dynamic. A snapshot cannot tell the whole story.

Glucose behaviour changes over time. Context from activity, meals, heart rate, and sleep-related signals helps interpret the current state.

Dynamic risk needs a dynamic view.

STATIC SNAPSHOT

102

mg/dL · one reading

Where are you now?

LIVING TIMELINE

Multiple signals.

Where might risk be moving?



THE VISION

What if a health profile could evolve with every new signal?

GlucoTwin combines a synthetic baseline profile with changing measurements to explore how estimated future glucose risk shifts as signals arrive.

CONCEPTUAL AND COMPUTATIONAL PROTOTYPE

Not a clinically validated replica of physiology.

BASELINE

Profile state

INCOMING

Changing signals



THE FLOW

From scattered signals to an evolving risk picture.

One complete flow from synthetic inputs to an interactive view.

01

Baseline profile

Personal context

02

Live-like synthetic signals

Changing measurements

03

Feature engineering

Signals become inputs

04

Risk prediction

Estimated event probability

05

Interactive dashboard

Replay the state

SYNTHETIC DATA · COMPUTATIONAL STATE · ESTIMATED GLUCOSE-RISK EVENT

THE STATE

A baseline profile meets a changing timeline.

All data shown is synthetic.

BASELINE PROFILE

Age · BMI · HbA1c · years since diagnosis · medication indicator

DYNAMIC SYNTHETIC STREAM

Glucose · activity · heart rate · sleep proxy · meal carbohydrate inputs

CHANGING COMPUTATIONAL STATE

Both streams combine into a changing computational state.

THE TECHNOLOGY

A transparent, reproducible machine-learning pipeline.

1
Synthetic data generation

2
Data preparation and feature engineering

3
Patient-level train/test split

4
Median imputation and feature scaling

5
Unweighted logistic regression

6
Risk probability

7
Streamlit dashboard

EARLIER CLASS-WEIGHTED LOGISTIC REGRESSION REMAINS AS A COMPARISON BASELINE.

THE DIFFERENCE

The value is in replaying change, not just displaying a score.

STATIC DISPLAY

1 score

One score at one moment.

GLUCOTWIN

Select · inspect · replay

Select a synthetic profile, inspect changing signals, and replay simulated states across a timeline.

No claim of autonomous physiological learning or a fully validated real-time digital twin.

THE PROTOTYPE

A concept you can interact with.

Illustrative interface mockup unless an actual application screenshot is supplied.

- Synthetic patient profile
- Glucose time-series chart
- Activity and heart-rate context
- Two-hour estimated risk
- Timeline replay control

PROJECT URL

https://twinpulsegit-ve5nzslj80h8.streamlit.app

REPLAY

Simulated timeline state

RISK

Two-hour estimate



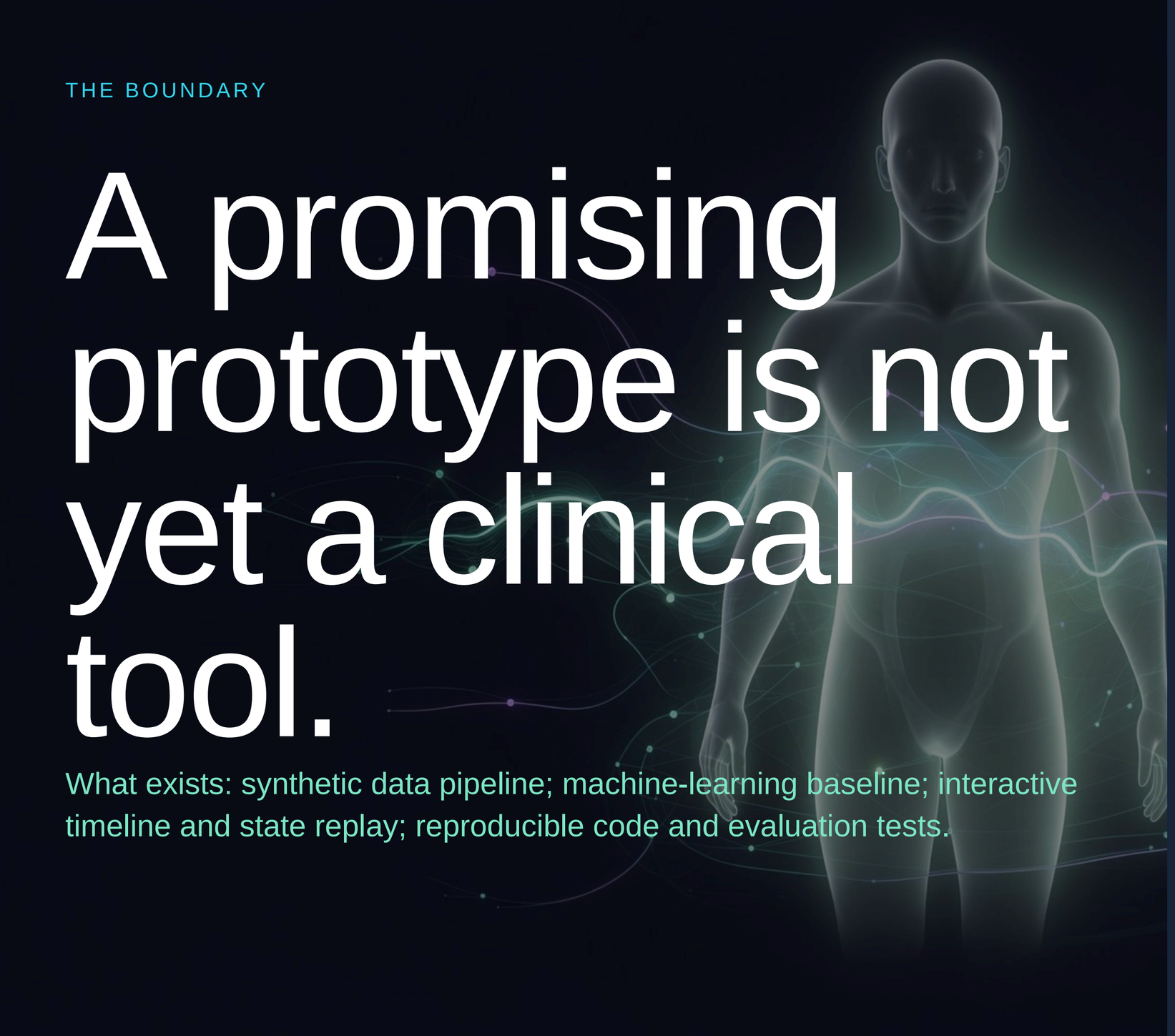
THE EVALUATION

Evaluation must account for rare events and patient-level separation.

Design choice	What it means
Synthetic cohort	80 profiles across five days at 15-minute intervals.
Patient-level split	60 training profiles and 20 held-out test patients.
Metrics	ROC-AUC, precision, recall, F1 score, and Brier score.
Dependence	Overlapping windows are not independent observations.
Engineering label	Whether maximum synthetic glucose in the next two hours reaches at least 180 mg/dL. Not a clinical diagnosis or emergency threshold.

THE BOUNDARY

A promising prototype is not yet a clinical tool.



What exists: synthetic data pipeline; machine-learning baseline; interactive timeline and state replay; reproducible code and evaluation tests.

WHAT REMAINS

Suitable real-world validation
Clinical and domain-expert review
External and prospective evaluation
Privacy, safety, bias, and calibration assessment

THANK YOU